

Animal Tissues – Exercise Answers

Question-wise answers based on the uploaded textbook exercise.

1. Choose the correct option

A. The study of structure and arrangement of tissue is called

Answer: Histology.

B. _____ gland is a gland which is both exocrine and endocrine.

Answer: Pancreas.

C. _____ cell junction allows intercellular passage of ions and small molecules.

Answer: Gap junction.

D. The protein found in cartilage is _____.

Answer: Chondrin.

E. Find the odd one out: Thyroid gland, Pituitary gland, Adrenal gland, Salivary gland.

Answer: Salivary gland — it is an exocrine gland, whereas the other three are endocrine glands.

F. Supriya stepped out into bright street light. Her eye pupil shrank. Identify the muscle responsible.

Answer: Circular smooth muscles of the iris (sphincter pupillae) are responsible.

2. Answer the following questions

A. Identify and name the type of tissues:

a. Inner lining of the intestine: Simple columnar epithelial tissue.

b. Heart wall: Cardiac muscle tissue.

c. Skin: Stratified squamous epithelial tissue.

d. Nerve cord: Nervous tissue.

e. Inner lining of the buccal cavity: Stratified squamous epithelial tissue.

B.

Answer: Animals in cold regions have a layer of fat below their skin because adipose tissue acts as a **thermal insulator** and reduces heat loss.

C.

Answer: The ear pinna can be folded and twisted because it contains **elastic cartilage**, which is flexible. Fingertips contain bones and therefore cannot be twisted in the same way.

D.

Answer: The tissue involved in the quick withdrawal from a hot plate is mainly **nervous tissue**, which carries the reflex impulse; skeletal muscles then contract to withdraw the hand.

E.

Answer: Long bones contain **bone marrow**, which is an important site of blood-cell formation. Injury to the marrow can reduce blood-cell production and may lead to anaemia.

F.

Answer: The shrinking of the pupil is caused by the **circular smooth muscles of the iris** (sphincter pupillae).

3. Answer the following questions

A. What is a cell junction? Describe different types of cell junctions.

Answer: A cell junction is a specialised connection between adjacent cells or between a cell and the extracellular matrix. Major types include **tight junctions** (seal neighbouring cells and prevent leakage), **adherens junctions** (help cells adhere through actin-associated proteins), **desmosomes** (strong cell-to-cell attachment), **hemidesmosomes** (attach cells to the basement membrane), and **gap junctions** (allow ions and small molecules to pass between neighbouring cells).

B. With the help of a neat labelled diagram, describe the structure of areolar connective tissue.

Answer: Areolar tissue is a loose connective tissue containing a soft matrix with **collagen fibres** and **elastic fibres**. It contains cells such as **fibroblasts**, **macrophages** and **mast cells**. It occurs beneath epithelia and around blood vessels and nerves, and helps bind and support tissues. **For the diagram, label:** fibroblast, collagen fibre, elastic fibre, mast cell, macrophage and ground substance.

C. Describe the structure of multipolar neuron.

Answer: A multipolar neuron has **one axon and many dendrites** arising from the cell body (cyton). The cell body contains a **nucleus** and cytoplasm. Dendrites receive impulses, while the axon carries impulses away from the cell body. The axon may be covered by a **myelin sheath** with gaps called **nodes of Ranvier**.

D. Distinguish between smooth muscles and skeletal muscles.

Answer: Smooth muscles are **involuntary**, usually spindle-shaped, and have a single nucleus; they occur in walls of organs such as the intestine and blood vessels. Skeletal muscles are **voluntary**, long and cylindrical, multinucleate and striated; they are attached mainly to bones and produce body movements.

4. Complete the following table

Cell / Tissue / Muscles	Functions
1. Cardiac muscles	Pump blood throughout the body.
2. Tendon	Connect skeletal muscles to bones.
3. Chondroblast cells	Form cartilage matrix and develop cartilage.
4. Mast cells	Secrete heparin and histamine.

5. Match the following and rewrite

'A' Group	'B' Group
1. Muscle	b. Sarcolemma
2. Bone	c. Periosteum
3. Nerve cell	d. Neurilemma
4. Cartilage	a. Perichondrium

Rewritten matches:

1. Muscle — Sarcolemma
2. Bone — Periosteum
3. Nerve cell — Neurilemma

4. Cartilage — Perichondrium

Note: The answers are written in exam-friendly form. For Question 3(B), draw a neat labelled areolar connective tissue diagram in your notebook/exam.